

30-Day Frontend System Design Interview Roadmap

Goal: Become interview-ready in Frontend System Design within 30 days.

WEEK 1 – Browser & Performance Foundations

Day 1: Rendering pipeline, DOM, CSSOM
Day 2: Critical Rendering Path, Reflow vs Repaint
Day 3: Event Loop (microtask vs macrotask)
Day 4: DevTools Performance analysis
Day 5: Network tab & waterfall analysis
Day 6: Memory leaks & React issues
Day 7: Mock explanation practice

WEEK 2 – Rendering Optimization (React Focused)

Day 8: Reconciliation & Virtual DOM
Day 9: React.memo, useMemo, useCallback
Day 10: State colocation & lifting state
Day 11: Keys & list rendering
Day 12: List virtualization
Day 13: Debounce & Throttle
Day 14: Practice – Why is your React app slow?

WEEK 3 – Network, Storage & Security

Day 15: HTTP lifecycle
Day 16: HTTP caching & headers
Day 17: CDN basics
Day 18: LocalStorage vs SessionStorage
Day 19: IndexedDB basics
Day 20: Auth token storage trade-offs
Day 21: Design a login system

WEEK 4 – Architecture & System Thinking

Day 22: MVC & Component architecture
Day 23: Feature-based folder structure
Day 24: Smart vs Dumb components
Day 25: State management patterns
Day 26: XSS & CSRF basics
Day 27: Design infinite scroll feed
Day 28: Design chat UI
Day 29: Design dashboard architecture
Day 30: Final mock interview practice

Daily Interview Rule:

- Is it scalable?
- Is it maintainable?
- Are re-renders optimized?
- Is network efficient?
- Are security risks handled?